

Varun Savita

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Date of Birth: 21 Sept 1995

Languages: English and Hindi

Address: Noida

Experienced professional excelling in Embedded and IoT development, and Cyber Security tools, with a solid track record in delivering end-to-end firmware solutions.

Technical Proficiency:

Programming Languages: C, Embedded C, Python, Bash Script.

IDE: Eclipse, Keil, Arduino, Vivado, Lab View, MPLAB.

MCU (IC): STM32I4, ATSAM3X, ATMEGA2560, nRF51822, ESP, PIC18F.

MCU (Boards): Arduino (Uno, Mega, Due, Pro Mini), Node MCU, NRF51-DK.

STM32 - Nucleo, Basys 3 Artix-7 FPGA, ESP32_wroom -32U, Beagle Bone Board.

SBC: Raspberry-Pi

Protocols (Wireless): HC-05, Wi-Fi, BLE-(Advertising, Uart), LoRa, LoRaWAN, GSM.

Protocols (Wired): SPI, I2C, UART.

Protocol (Application): MQTT

Network Server: TCL, ChirpStack, The Things Network.

LoRaWAN Gateway: Kerlink-ifemtocell, RAK-7240, Laird -RG186, RAK2245 Pi Hat.

Simulation: Model SIM, Xilinx ISE, System Generator, Proteus, MATLAB.

Database: MongoDB

Web Technology: HTML

Management tools: JIRA and Confluence

Code Manage: GitHub.

CI/CD: Jenkins

Cloud: AWS-EC2, S3.

Operating Systems- Window, Linux

Cyber security: Coverity, Blackbuck, Polaris, SonarQube, CVSS 4.0.

Design - Microsoft Threat Modelling Design

Packet Analysis – Wireshark.

Experience:

Motherson Technology Services Limited. Noida, UP – (May 2022)

I am currently working as a secure developer for a **Client (Carrier HRDC)**. In this role, I am responsible for DevSecOps activities, managing cyber security tools, and providing support to the engineering team. My duties involve utilizing SAST/SCA scanning tools like Coverity, Blackbuck, Polaris, BDBA, and SonarQube. Additionally, I am responsible for maintaining these tools and collaborating with the architect to review the reports.

- Written Python script for Automation.
- Worked on Linux and Windows systems.

Ezytech Industries - (July 2020 to May 2022): Embedded Software Developer

I worked on end-to-end development in the CIJ printer project.

Project – CIJ Printing Machine

- Responsible for Firmware, Architecture, and testing.
- Environment/Tools: PIC18F722, Arduino Due, LCD240*128, SRAM, keypad with 70 keys, GPIO expander, solenoid valve, IR sensor.

Icapotech Pvt. Ltd - (Feb 2020 to June 2020): Embedded and IoT Developer

I work as a Firmware and Architecture developer at Icapotech. I primarily handle firmware and set-up cloud with controller for the internal R & D projects.

Project – Home Automation

- Responsible for Firmware and Architecture.
- Environment/Tools: ESP IDF, ESP32, MQTT, AWS.

Project – IR- Blaster

- Responsible for Front-End (Node RED), Architecture, and cloud set-up.
- Environment/Tools: Arduino, ESP8266, MQTT Cloud, AWS- EC2, Node-Red, MongoDB.

DSPWorks India Pvt. Ltd. - (Oct 2019 to Jan 2020): Developer Member

My contribution was in Firmware for LoRaWAN node and server project.

Project – Vehicle Tracker using GPS and LoRaWAN

Responsible:

- Firmware on STM32L4 and ESP
- Setup ChirpStack LoRaWAN Network Server on AWS and DigitalOcean.
- Set up LoRaWAN Gateway.
- Testing LoRaWAN Device and Analysis on TCL and ChirpStack Server.

Project – Barcode Scanner

Read the barcode using a scanner and send server then Identify our product.

Environment/tools: GPS(Quectel-L70R), LoRaWAN-1.0.3(LoRa Stack 1.3.1), Server (TCL and ChirpStack), LoRaWAN Gateway (RAK833), ESP-Wroom-02U/ ESP32_wroom -32U, Wave share -SKU:14810 barcode scanner module.

Smartify Health Pvt. Ltd - (Feb 2018 to Sept 2019): Embedded and IoT Developer

I worked as an Embedded and IoT engineer in Smartify. My primary responsibility here was firmware and interface sensor with controller for the internal R & D projects as well as client projects.

Project – Smart Parking in BPCL

The project is about Smart Parking based on IoT, an application is made which is Android, Dashboard, and Display supported. To access Data which is stored on AWS via LoRa Server.

- Responsible for Hardware Team Lead.
- Preparing resource planning.
- Requirements analysis, task allocations & tracking.
- Review and tracking of the material deliverables.
- Interacting with clients and coordinating the team.

Environment/Tools: Wireless geomagnetic parking sensor, LoRaWan Gateway, LoRa Server & TTN,

BLE (nRF51822), LoRaWAN Module, GSM900A, Display(P10).

Project – Smart IV

Smart IV monitors are an innovation in the healthcare sector that helps nurses and doctors provide special attention to patients. Additionally, provides data to patients' family members through an app.

- LCD UI Developer
- Responsible for Tech Specs
- Work with controller (ATMEGA 2560, ATSAM3X)

Project – Nurse Calling

- Monitoring patient parameters (Temp, Heart rate, falling data, nurse call button) on a mobile app.
- Hardware - BLE (nRF51822 and Sensor tag IC).

Project – Smart Parking

- LoRaWAN - LM130H1
- Gateway - Laird - Sentrius RG168
- Parking Sensor - D0100

SocialBuzz Technology Pvt. Ltd (July 2017 to Jan 2018):

Responsibilities as an intern (Stipend):

- Programming in C and Python languages.
- To work on various protocols like UART, I2C, and SPI to send data to software end from the microcontroller.
- To work on various Embedded Development Boards such as Arduino, ESP8266, and Raspberry Pi core Devices.
- To work on a new and innovative idea, design & Development of such ideas.
Design, debugging, and development of the product which are based on recent technologies.

Projects:

- Hexaware project (Cycle Energy Meter, Tree of Hope).
- Tweet cafe.
- Selfie Mirror
- Seagate Project (Video Display)
- Big Bazaar (Ping Button and Like Dislike Feedback)

Environment/Tools: NODE-MCU. Arduino Mega2560, 7 Segment, IR sensor, Exercise Cycle, RGB Led strip. Raspberry Pi, RTC1307, Electric Magnet, Capacitive touch sensor, USB to TTL, LCD, RFID –RC522, NODE-MCU, MQTT protocol.

Other Experience:

Summer Intern:

- Research Designs & Standards organization, Lucknow. (3 July- 3 aug16)
- Hindustan Aeronautics Limited (HAL). (1 June 16 -30 June 1)

Trainee/ Workshops/ Certification:

- Secure Development by PurpleSynapz™ (2022).
- CETPA, Lucknow – 4 weeks summer trainee of VHDL Language after 4th-sem.

- Design and simulation of a seven-floor elevator controller based on VHDL Technology.
- Design and simulation of a four-way traffic light controller using Finite State Machine Method based on VHDL technology.
- Printed Circuit Board and circuit Designing organized by IEEE.
- Embedded System organized by Electronics and ICT Academy by Dhananjay V. Gadre (NSIT), 5 days.

B-tech Project:

- Design and simulation of time watch (23:59:59) based on VHDL Technology.
- Implementation of Signal Processing Algorithms for Radar Based Subsurface Imaging on Artix-7 FPGA.

Objective:

In this academic project, under the guidance of Prof. Puli Kishore Kumar, the main objective is to implement signal processing algorithms for the detection and location of buried objects. Two algorithms were used for this purpose, Kirchhoff Migration Algorithm and Back Projection Algorithms using wideband frequency, MATLAB, and SYSTEM GENERATOR.

Land mine detection, Identifying the objects buried under rubble and foliage, Assisting first responders during military operations.

- An Artificial Neural Network Model for Dual Frequency Microstrip Antenna and its Implementation in FPGA.

Design of Microstrip Antenna using Coaxial Feeding Technique in Computer simulation Technology software and Generate Neural Network modeling in MATLAB using nntool.

- Four-way traffic light controller based on digital IC on 4th-semester design circuit using CMOS IC (4017) and Timer 555.

Academics:

- 2013-2017 B-Tech (ECE): **National Institute of Technology Patna** – CGPA - 6.78
- 2010-2012 - 12 standard: Chacha Nehru Smarak Inter College -84.8 %
- 2009 -2010 - 10 standard: Chacha Nehru Smarak Inter College - 66.67 %

Extra-Curricular Activities:

- Event co-coordinator in robotics CORONA -2016
- 3 prizes in robotics CORONA -2015
- Member of the National Service Scheme
- Member of Uttar Pradesh Bharat Scouts and Guide.

Strengths:

- Working under pressure.
- Strong leadership skills.
- Quick learner and flair to learn new things.

Declaration:

I hereby declare that all the information mentioned above is true to my knowledge and I bear the responsibility for the above-mentioned particulars.

Place – Kanpur

Varun Savita